Interpreting Data

Exam code: 0620 & 0971

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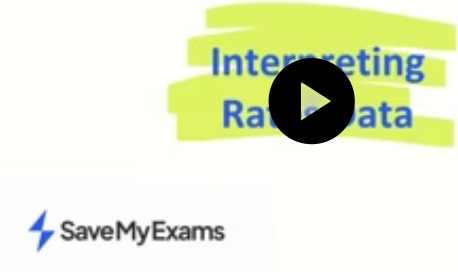

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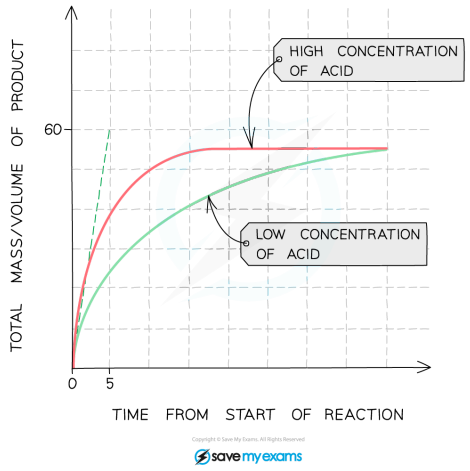
Interpreting data

- Data recorded in rate studies is used to **plot graphs** to calculate the rate of a reaction
- Plotting a graph until the completion of the reaction shows how the rate of reaction changes with time
- The rate of reaction is always quickest at the start of the chemical reaction
  - This is when there are the most reactants available to collide and form products
  - On a graph, the line at the start of the reaction has the steepest gradient
- As the reaction progresses, the rate of reaction decreases
  - This is because the concentration of reactants decreases as they are being used up
  - On a graph, the line becomes less steep
- The reaction eventually stops
  - This is because at least one of the reactants has been used up
  - The rate of reaction is now zero
  - On a graph, this is when the line becomes horizontal
- The amount of product formed in a reaction is determined by the limiting reactant:
  - If the amount of limiting reactant increases, the amount of product formed increases

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

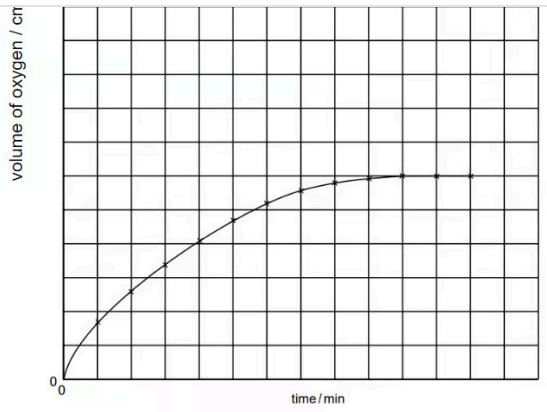
graph making it easier to see how the variable influences the rate
    - For example, plotting the effect of concentration on a reaction between the acid and marble chips



- Drawing a **tangent** to the slope allows you to show the gradient at any point on the curve
    - A steeper slope means a quicker reaction and a higher rate of reaction
- The volume of a gaseous product would increase to a maximum over time
    - So, the line levels out indicating the reaction is over
    - Since the volume and mass would be proportional, this could also be a graph of the mass of product versus time

Worked Example

0.2 g of manganese(IV) oxide was added to 25 cm^3 of 0.1 mol / dm^3 hydrogen peroxide solution. The volume of oxygen produced every minute was recorded and the results are shown on the graph.

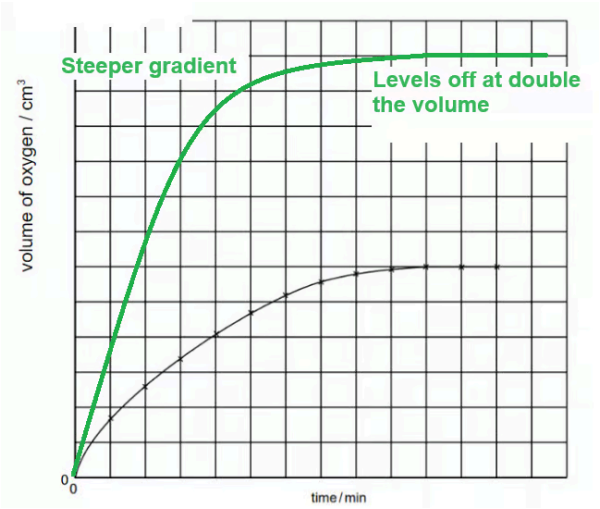


The experiment was repeated using the same mass of manganese(IV) oxide and at the same temperature but using 25 cm^3 of 0.2 mol / dm^3 of hydrogen peroxide solution.

Sketch the curve for the results of this experiment on the same grid.

Answer:

1. Deduce how the initial gradient will be different from the original graph
   - The hydrogen peroxide solution is twice as concentrated
   - So, the rate of reaction will be greater and the initial **gradient will be steeper**
2. Deduce how much product will be formed compared to the original experiment
   - The amount of hydrogen peroxide determines the amount of oxygen produced
   - In the 2nd experiment, there are twice as many hydrogen peroxide molecules in the same volume
   - So, this will produce **double** the amount of oxygen **in the same time**
3. Sketch the graph



particular point

- To calculate the rate of reaction at any given time / point in a chemical reaction, you need to find the **gradient of the curve** at that time / point
- To do this:
  - Draw a **tangent to the curve**
  - Calculate the change in the y-axis
  - Calculate the change in the x-axis
  - Calculate the gradient of the tangent using:
    - Rate of reaction (or gradient) = $\frac{\text{change in y}}{\text{change in x}}$

Worked Example

Iodine and methanoic acid react in aqueous solution.

$$I_2\,(aq) + HCOOH\,(aq) \rightarrow 2I^-\,(aq) + 2H^+\,(aq) + CO_2\,(g)$$

The rate of reaction can be found by measuring the volume of carbon dioxide produced per unit time and plotting a graph as shown:

Calculate the rate of reaction at 20 seconds.

Answer:

- Draw a tangent to the curve at 20 seconds:

- Complete the triangle and read off the values of x and y
- Determine the rate of reaction / gradient of the line:
  - Rate of reaction (or gradient) = $\frac{\text{change in y}}{\text{change in x}}$
  - Rate of reaction (or gradient) = $\frac{24}{40}$ = 0.60 cm^3 / s

Examiner Tips and Tricks

If the amount of reactant used up is being monitored, then the graph will **fall** with the steepest gradient at the start, becoming less steep until it levels off to a horizontal line.

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